TOPIC 03 · INTERACTIVE DASHBOARDS WITH COPILOT, SKILLS AND RULES
LAB 8 THEME · DASHBOARD SKILL, REFRESH & SHARE

Lab 8 — Turn the Dashboard into a Reusable Skill, Refresh It and Share It Safely

Maps to	LO3: use Copilot to write a dashboard skill, run it on a workforce workbook with rules, prove the dashboard refreshes with new rows, and check it is safe to share
Duration	45 minutes
Tools	Excel · Copilot (Chat only, Plan, Allow editing) · .Rules · custom skill · PivotTables, slicers, refresh
Mock data	data/BrightPath_Workforce_Headcount.xlsx
Lab files	rules-template.txt (the .Rules text to paste into column A); skills/hr-workforce-dashboard/SKILL.md (ready-made skill to upload if you run short of time)
Course	Microsoft 365 Copilot for Excel (C34) · v3.0
Prompts	https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-8

Goal

Build BrightPath's workforce and attrition dashboard from a one-line skill call, prove it updates when new joiners arrive, and make sure no small group can be identified.

What You Will Do

BrightPath's HR Director needs the same kind of dashboard every month. Instead of repeating a long brief, you will ask Copilot to write an hr-workforce-dashboard skill, save it to OneDrive and run it on the workforce workbook, whose .Rules sheet holds the period, the attrition definition and the privacy rule. Then you will append eight July joiners and refresh to prove the dashboard is live, check that no slicer combination exposes fewer than 5 people, and ask Copilot for a short leadership summary.

What You Will Build

An hr-workforce-dashboard skill in OneDrive, and BrightPath_Workforce_Headcount.xlsx with a .Rules sheet, helper columns, a Checks sheet and an HR_Dashboard with six KPI cards, four charts and Location and Job_Level slicers - refresh-tested and privacy-checked

Prerequisites

- Lab 6 and Lab 7 completed: you can save a custom skill to OneDrive and build a dashboard with rules.
- This lab folder's data workbook, rules-template.txt and skills/hr-workforce-dashboard/SKILL.md (fallback).

Data note

Use only the supplied fictional BrightPath data or information you are authorised to process. Do not paste credentials, confidential business material or real employees' personal data into Copilot, rules or skills.

Steps

Step 1. Go to onedrive.com (or your SharePoint site), open or create the folder work-c34 and select Upload > Files. Upload BrightPath_Workforce_Headcount.xlsx from this lab's data folder, then open it in Excel for the web or select Open in Desktop App. OneDrive gives you AutoSave and version history (recent Excel builds can also run Copilot on local files). Check that Formulas > Calculation Options is Automatic, then open Copilot from the Copilot button on the Home tab or the floating Copilot icon at the bottom right. Read the About_This_Data sheet before you prompt. Do not touch the New_Joiners_Jul2026 sheet until the refresh test.

PROMPT / CHECK — copy and adapt

```
Table : tblEmployees (420 employees hired since 2019; Status Active or Exited)
Period: 2025-07-01 to 2026-06-30
Attrition rate = exits in period / ((active on 2025-07-01 + active on 2026-06-30) / 2)
```

Step 2. Create the workbook rules: + (Add work content) > Create workbook rules, then paste rules-template.txt into .Rules!A1 or send the prompt below. Note the privacy rule - the skill will follow it.

PROMPT / CHECK — copy and adapt

```
ALLOW EDITING:
On the .Rules sheet, replace everything in column A with these rules, one per cell from A1 down.
Keep each ## heading in its own cell. Do not change any other sheet.
## WORKFORCE DASHBOARD STANDARDS
Never change or delete existing rows of tblEmployees.
Never show Employee_ID, Monthly_Salary or any individual record on a dashboard.
The 12-month period is 2025-07-01 to 2026-06-30. Active on a date = Hire_Date on or before it and Exit_Date blank or after it.
Active headcount = employees with Status Active now, so new joiners count as soon as they are added to tblEmployees.
Attrition rate = exits in the period / ((active on 2025-07-01 + active on 2026-06-30) / 2).
New joiners after 2026-06-30 do not change it.
Tenure_Years = years from Hire_Date to Exit_Date, or to 2026-06-30 if still active; use 0 for anyone hired after 2026-06-30.
Do not show any group of fewer than 5 employees. Group it into Other or leave it out.
## LAYOUT
Dashboard sheet HR_Dashboard: title in A1, "Data as at 2026-06-30" in A2, KPI cards in row 4, charts in a grid below, slicers on the right.
Put every PivotTable on a sheet named Dash_Data. Never merge cells. Hide gridlines on HR_Dashboard.
## VISUAL STYLE
Colours: navy #1F4E79 for headcount, teal #2A9D8F for hires, amber #F4A261 for exits and attrition.
Every chart title states the finding. Rates as % with 1 decimal. Years with 1 decimal.
```

Step 3. Let Copilot write the skill. In Chat only, describe the recipe you followed in Lab 7, made general. Read the reply and ask for changes until the steps are clear.

PROMPT / CHECK — copy and adapt

```
CHAT ONLY:
Write a complete SKILL.md for a Copilot in Excel skill named hr-workforce-dashboard.
Description: Use when I ask for a workforce, headcount or attrition dashboard from an employee table.
Steps:
1) Read the .Rules sheet first and follow it.
2) Add helper columns Active_Start, Exit_In_Period and Tenure_Years using the period in the rules.
3) Create a Checks sheet with independent COUNTIFS formulas for active headcount (Status Active), headcount at period start, headcount at period end, exits, voluntary exits, attrition rate and hires.
4) Put all PivotTables on Dash_Data.
5) Build HR_Dashboard: six KPI cards (Active headcount, Exits, Attrition rate, Voluntary attrition rate,
```

```
   Avg tenure of active staff, Hires), charts of headcount by Department, attrition rate by
Department, hires vs exits by quarter and headcount by Location, and slicers for Location and
Job_Level.
6) Compare every KPI card with Checks and report any difference.
7) List any slicer combination with fewer than 5 employees.
Return only the SKILL.md text, starting with the --- front matter.
```

Step 4. Save it as before: copy the reply into a plain-text file named SKILL.md, create the folder hr-workforce-dashboard in your OneDrive skills folder and upload the file (or upload the ready-made one from this lab folder). In Excel select Settings (...) > Manage plugins & skills > Custom skills > Refresh and turn the skill on.

PROMPT / CHECK — copy and adapt

```
OneDrive : .../skills/hr-workforce-dashboard/SKILL.md
Check : folder name = name: line = hr-workforce-dashboard
```

Step 5. Run the skill in Plan mode: one line replaces the long brief, because the skill holds the steps and the rules hold the definitions. Review the plan, approve it, and select Done when Copilot finishes.

PROMPT / CHECK — copy and adapt

```
PLAN MODE:
@hr-workforce-dashboard Build the workforce dashboard for the 12 months ending 2026-06-30.
```

Step 6. Verify. Ask Copilot to compare the KPI cards with the Checks sheet the skill built, and to show the attrition formula. Compare both with the Test It values - a skill can build a wrong check too.

PROMPT / CHECK — copy and adapt

```
CHAT ONLY:
Compare each KPI card on HR_Dashboard with the Checks sheet and tell me which, if any, do not
match.
Show the formula behind the Attrition rate card.

ALLOW EDITING (if needed):
The attrition rate must use average headcount = (active on 2025-07-01 + active on 2026-06-30)
/ 2, as the rules say.
```

Step 7. Refresh test: ask Copilot to append the 8 rows from New_Joiners_Jul2026 to tblEmployees and refresh every PivotTable. Every KPI and chart that counts active staff must update.

PROMPT / CHECK — copy and adapt

```
ALLOW EDITING:
Copy the 8 data rows in New_Joiners_Jul2026!A2:L9 to the first empty rows directly below
tblEmployees so the table expands. Then refresh all PivotTables in the workbook and tell me
the new Active headcount and the number of rows in tblEmployees.

Expected after refresh:
Active headcount : 237 -> 245
tblEmployees rows: 420 -> 428
New joiners by department: Recruitment +2, Sales & Client Services +2, IT +4
```

Step 8. Check it is safe to share, then ask for a leadership summary that uses only dashboard numbers. Copilot drafts; you check every number before it leaves HR. Log your prompts.

PROMPT / CHECK — copy and adapt

```
CHAT ONLY:
Review HR_Dashboard before I share it with the leadership team. Does any element expose
individual employee data or a group smaller than 5? Is each KPI labelled with its period and
definition?

CHAT ONLY:
Write a 5-line summary of HR_Dashboard for BrightPath's leadership team. Use only numbers
shown on the dashboard, and label any explanation as a hypothesis to test.

LOG (Allow editing):
Prompt_Log already lists every prompt in this lab. For each one I ran, fill in What Copilot
did and How I verified, and set Decision to Keep. I will tell you any row to change to Revise
or Undo.
```

Sample Prompts

Use these prompts in the Copilot pane (or adapt them). Name the table and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

Copy-ready prompts online:
https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-8

#	Purpose	Prompt
1	Write a skill	Write a SKILL.md named hr-workforce-dashboard that reads the .Rules sheet first and builds a verified workforce dashboard.
2	Run a skill	@hr-workforce-dashboard Build the workforce dashboard for the 12 months ending 2026-06-30.
3	Refresh	I added rows to tblEmployees. Refresh all PivotTables and tell me which KPI values changed.
4	Privacy	Which slicer combinations show a group of fewer than 5 employees? Suggest how to suppress them.
5	Explain	Explain how the Attrition rate card is calculated, step by step.
6	Improve the skill	Update the hr-workforce-dashboard SKILL.md so it also adds a timeline on Hire_Date. Return the full file.
7	Summary	Write a 5-line leadership summary using only the numbers on HR_Dashboard.

Test It

Check	Expected result
Active headcount	237 before the refresh
Exits and attrition	29 exits (25 voluntary); 13.0% attrition
Hires and tenure	56 hires; average tenure about 3.1 years
Highest-attrition department	Recruitment: 21.8% (12 exits)
Refresh test	After adding the July joiners: headcount 245, 428 rows
Privacy	No Employee_ID, salary, individual record or group under 5 on the dashboard

Checkpoint

You now have two custom skills in OneDrive. Run hr-workforce-dashboard again next month on new data, and improve the skill whenever you correct Copilot twice.

Troubleshooting

If this happens	Fix
The new rows did not join the table and nothing changed on refresh.	Ask Copilot to resize tblEmployees to include the 8 new rows, then refresh all PivotTables again.
Attrition shows 12.2% (29 / 237) instead of 13.0%.	The card divides by closing headcount only. Ask Copilot to use the average of opening and closing headcount, then add that wording to the skill.
The skill ignores the .Rules sheet.	Check the sheet is named .Rules and is visible, and that step 1 of the skill says to read it. Start a New chat.
Helper columns show #VALUE! for blank Exit_Date.	Ask Copilot to treat blank Exit_Date explicitly, e.g. OR([@Exit_Date]="",[@Exit_Date]>DATE(2025,7,1)).

Challenge

Improve the skill so it also adds a Timeline on Hire_Date and a chart of average Engagement_Score by Department for active staff. Save, refresh and rerun it, then check one department's average with AVERAGEIFS.

Reflection

What would happen to trust in this dashboard if one leader could filter it down to a single named team member?